

PROBABILITY & STATISTICS

Final Project



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Introduction

This project investigates what factors drive the online popularity of news articles, measured by the number of social media shares. Using the UCI Online News Popularity Dataset (39,644 articles, 61 features), we applied statistical analysis, regression modeling, classification, and hypothesis testing to understand and predict article virality.

Dataset Description

The dataset contains 39,644 articles from Mashable with features covering content structure (word counts, number of images, videos, hyperlinks), keyword statistics, publication timing (day of week, weekend), LDA topic scores, and sentiment metrics (subjectivity, polarity). The target variable is shares, representing social media shares. Key descriptive statistics show the mean shares value is 3,395 with a standard deviation of 11,627, indicating extreme right skew. The minimum is 1 share and the maximum is 843,300.

Data Loading and Initial Inspection

The OnlineNewsPopularity.csv dataset was loaded into a pandas DataFrame. We started by inspecting data types and checking for missing values using `df.info()`, then generated descriptive statistics with `df.describe()` to get a feel for the overall structure and distribution. Column names were also cleaned to remove any extra whitespace.

```
3
4 #displaying the first 5 rows
5 #print(df.head()) for more formatted output im using display instead of print func
6 display(df.head())
```

	url	timedelta	n_tokens_title	n_tokens_content	n_unique_tokens	n_non_st
0	http://mashable.com/2013/01/07/amazon-instant-...	731.0	12.0	219.0	0.663594	
1	http://mashable.com/2013/01/07/ap-samsung-spon...	731.0	9.0	255.0	0.604743	
2	http://mashable.com/2013/01/07/apple-40-billio...	731.0	9.0	211.0	0.575130	
3	http://mashable.com/2013/01/07/astronaut-notre...	731.0	9.0	531.0	0.503788	
4	http://mashable.com/2013/01/07/att-u-verse-apps/	731.0	13.0	1072.0	0.415646	

5 rows × 61 columns

```
1 # Display concise summary of the DataFrame
2 df.info()

5    n_non_stop_words      39644 non-null float64
6    n_non_stop_unique_tokens  39644 non-null float64
7    num_hrefs            39644 non-null float64
8    num_self_hrefs       39644 non-null float64
9    num_imgs             39644 non-null float64
10   num_videos           39644 non-null float64
11   average_token_length  39644 non-null float64
12   num_keywords         39644 non-null float64
13   data_channel_is_lifestyle  39644 non-null float64
14   data_channel_is_entertainment  39644 non-null float64
15   data_channel_is_bus       39644 non-null float64
16   data_channel_is_socmed    39644 non-null float64
17   data_channel_is_tech      39644 non-null float64
18   data_channel_is_world     39644 non-null float64
```

```
1 # Generate descriptive statistics
2 display(df.describe())
```

	timedelta	n_tokens_title	n_tokens_content	n_unique_tokens	n_non_stop_words	n_non_stop_unique_t
count	39644.000000	39644.000000	39644.000000	39644.000000	39644.000000	39644.0
mean	354.530471	10.398749	546.514731	0.548216	0.996469	0.4
std	214.163767	2.114037	471.107508	3.520708	5.231231	3.2
min	8.000000	2.000000	0.000000	0.000000	0.000000	0.0
25%	164.000000	9.000000	246.000000	0.470870	1.000000	0.4
50%	339.000000	10.000000	409.000000	0.539226	1.000000	0.4
75%	542.000000	12.000000	716.000000	0.608696	1.000000	0.7
max	731.000000	23.000000	8474.000000	701.000000	1042.000000	650.0

8 rows × 60 columns

Exploratory Data Analysis (EDA)

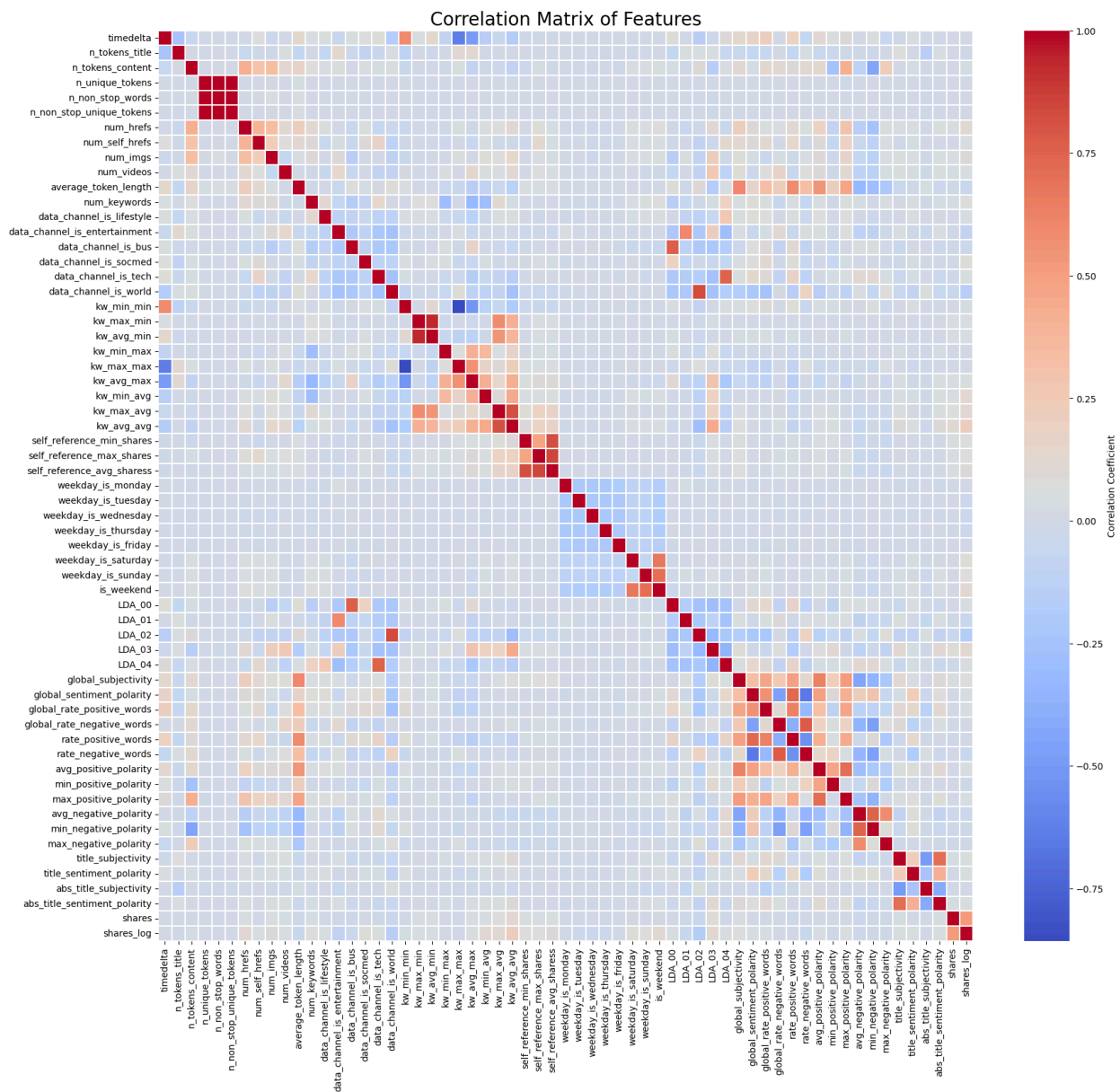
Target Variable Distribution

The shares variable is highly right-skewed, with most articles receiving fewer than 5,000 shares while a small number go viral. Only 203 articles exceeded 50,000 shares, and only 5 exceeded 500,000. The skewness was addressed by applying a log transformation ($\log(\text{shares} + 1)$), which produced a near-normal distribution with mean 7.48 and standard deviation 0.93.

Correlation Analysis

The features most positively correlated with log-transformed shares were `is_weekend` (0.156), `kw_avg_avg` (0.151), `data_channel_is_socmed` (0.120), and `data_channel_is_tech` (0.117). The most negatively correlated were `data_channel_is_world` (-0.150) and `LDA_02` (-0.150).

Explaining Unusual Correlations: The strong positive correlation of `is_weekend` likely reflects reader behavior ; weekend audiences browse more leisurely and are more inclined to share content. The negative correlation of `data_channel_is_world` (world/international news) may reflect that hard news content is consumed but shared less than entertainment or social content. The high correlation of `kw_avg_avg` suggests that articles targeting already-popular keyword niches inherit their shareability.



Redundant Features

Three feature groups showed near-perfect multicollinearity (correlation > 0.9): `n_unique_tokens`, `n_non_stop_words`, and `n_non_stop_unique_tokens` were all correlated at $r = 1.00$. Additionally, `kw_max_min` and `kw_avg_min` had $r = 0.94$. Features `n_non_stop_words` and `n_non_stop_unique_tokens` were dropped.

Encoding

All categorical variables in the dataset were already binary-encoded as 0/1 dummy variables, so no additional encoding was required.

Data Preprocessing

Dropping Redundant Features

Based on the correlation findings, `n_non_stop_words` and `n_non_stop_unique_tokens` were removed from the dataset to reduce redundancy.

Advanced Multicollinearity Check Using VIF

We calculated Variance Inflation Factor (VIF) scores for all remaining features. Those exceeding a threshold of 5 or 10 were considered problematic. As a result, `weekday_is_thursday`, `is_weekend`, `weekday_is_monday`, the LDA topic features (LDA_00 through LDA_04), and `n_unique_tokens` were all removed due to high multicollinearity.

	feature	VIF
36	rate_positive_words	18.279236
37	rate_negative_words	16.204725
26	self_reference_avg_sharess	16.142684
16	kw_max_min	11.381786
7	average_token_length	11.225688
17	kw_avg_min	10.983352
23	kw_avg_avg	10.642090
25	self_reference_max_shares	8.140008
33	global_sentiment_polarity	7.777912
41	avg_negative_polarity	7.669660
22	kw_max_avg	7.498614
35	global_rate_negative_words	6.640622
38	avg_positive_polarity	6.047222
42	min_negative_polarity	5.236790
24	self_reference_min_shares	5.163559
19	kw_max_max	4.690294
34	global_rate_positive_words	4.667091
14	data_channel_is_world	3.837248
15	kw_min_min	3.804455
20	kw_avg_max	3.782924
40	max_positive_polarity	3.335471
13	data_channel_is_tech	3.023069
32	global_subjectivity	2.964457
43	max_negative_polarity	2.936290
11	data_channel_is_bus	2.683826
10	data_channel_is_entertainment	2.613015
47	abs_title_sentiment_polarity	2.388647
44	title_subjectivity	2.350909
2	n_tokens_content	2.231197

21	kw_min_avg	2.190813
0	timedelta	2.089572
39	min_positive_polarity	1.909792
12	data_channel_is_socmed	1.685634
3	num_hrefs	1.674788
9	data_channel_is_lifestyle	1.555377
8	num_keywords	1.493990
5	num_imgs	1.418317
46	abs_title_subjectivity	1.412395
4	num_self_hrefs	1.407481
18	kw_min_max	1.397584
45	title_sentiment_polarity	1.323758
28	weekday_is_wednesday	1.246094
27	weekday_is_tuesday	1.245711
29	weekday_is_friday	1.209719
6	num_videos	1.201891
1	n_tokens_title	1.140093
31	weekday_is_sunday	1.130577
30	weekday_is_saturday	1.117279

Splitting Data into Training and Testing Sets

The cleaned dataset was divided into features (X) and the target variable (y = shares_log). The url column, the original shares column, shares_log itself, and all previously removed features were excluded from X. We then performed an 80/20 train-test split.

```
Shape of X_train: (31715, 48)
Shape of X_test: (7929, 48)
Shape of y_train: (31715,)
Shape of y_test: (7929,)
```

Feature Scaling (Standardization)

We applied StandardScaler to normalize numerical features so that no single feature dominates the model due to its scale. The scaler was fitted on the training data and then used to transform both the training and test sets.

```
X_train_scaled shape: (31715, 48)
X_test_scaled shape: (7929, 48)

Descriptive statistics for X_train_scaled (first 5 columns):
```

	timedelta	n_tokens_title	n_tokens_content	num_hrefs	num_self_hrefs
count	3.171500e+04	3.171500e+04	3.171500e+04	3.171500e+04	3.171500e+04
mean	-8.065439e-17	-2.215755e-16	-8.110247e-17	3.091751e-17	-1.523472e-17
std	1.000016e+00	1.000016e+00	1.000016e+00	1.000016e+00	1.000016e+00
min	-1.617380e+00	-3.961022e+00	-1.162062e+00	-9.617182e-01	-8.618368e-01
25%	-8.882217e-01	-6.612231e-01	-6.405094e-01	-6.085842e-01	-6.003881e-01
50%	-7.025525e-02	-1.898233e-01	-2.921035e-01	-2.554501e-01	-7.749071e-02
75%	8.739117e-01	7.529762e-01	3.582541e-01	2.742509e-01	1.839580e-01
max	1.761990e+00	5.938374e+00	1.449086e+01	2.587647e+01	2.946621e+01

Linear Regression Model

Training and Interpretation

A Linear Regression model was trained on the scaled training data. We then examined the resulting coefficients to understand how each feature influences the log-transformed share count.

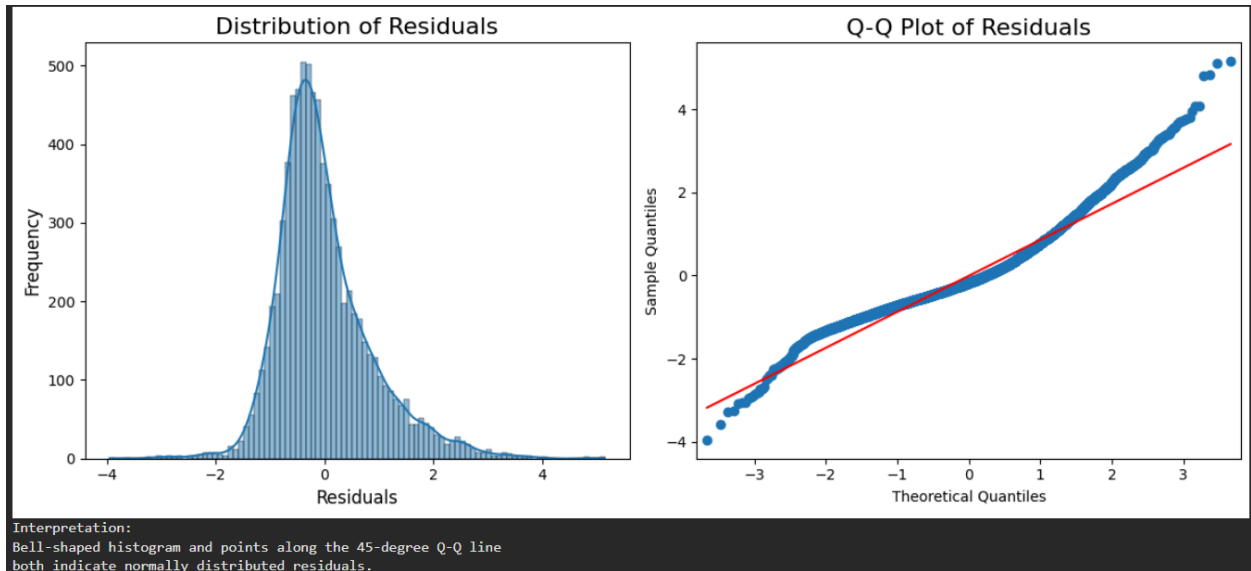
```
Linear Regression Model Trained Successfully
Model Intercept: 7.477649836210371
```

Model Coefficients:	
	Coefficient
kw_avg_avg	0.444476
data_channel_is_tech	0.079018
kw_max_min	0.070808
data_channel_is_socmed	0.068578
weekday_is_saturday	0.062814
weekday_is_sunday	0.061262
kw_min_min	0.060833
rate_positive_words	0.049471
global_subjectivity	0.049258
num_hrefs	0.048616
abs_title_subjectivity	0.029295
self_reference_avg_sharess	0.026956
self_reference_min_shares	0.026636
num_imgs	0.025864
num_keywords	0.025679
n_tokens_content	0.023701
data_channel_is_bus	0.021343
title_subjectivity	0.021298
n_tokens_title	0.019520
title_sentiment_polarity	0.019323
timedelta	0.016366
weekday_is_friday	0.015245
rate_negative_words	0.014099
abs_title_sentiment_polarity	0.010413
global_rate_negative_words	0.009679
num_videos	0.009476
kw_max_max	0.008953
max_negative_polarity	0.005893
min_negative_polarity	0.005311
self_reference_max_shares	0.004136
data_channel_is_lifestyle	0.001361
global_sentiment_polarity	-0.002242
avg_positive_polarity	-0.003442
max_positive_polarity	-0.008848
weekday_is_wednesday	-0.015940
min_positive_polarity	-0.017615
weekday_is_tuesday	-0.017929
avg_negative_polarity	-0.021680
global_rate_positive_words	-0.022480
kw_min_max	-0.024924
data_channel_is_world	-0.032187
num_self_hrefs	-0.032618
kw_avg_max	-0.034529
kw_min_avg	-0.059547
data_channel_is_entertainment	-0.067659
average_token_length	-0.074168
kw_avg_min	-0.083744
kw_max_avg	-0.262132

Checking Assumptions

Homoscedasticity: A residuals vs. predicted values plot was generated to check for constant variance. The plot revealed some signs of heteroscedasticity.

Normality of Residuals: A histogram and Q-Q plot of the residuals were produced. Both indicated some deviation from a perfectly normal distribution.



Model Evaluation

Predictions were made on the test set and evaluated using MAE, MSE, RMSE, and R-squared (R^2).

```
--- Linear Regression Model Performance ---  
Mean Absolute Error (MAE): 0.6462  
Mean Squared Error (MSE): 0.7510  
Root Mean Squared Error (RMSE): 0.8666  
R-squared (R2): 0.1240
```

Comparison Against a Mean Baseline

To put the results in context, we compared the linear regression model against a simple mean baseline. The regression model meaningfully outperformed the baseline, achieving a positive R^2 of 0.1240 compared to the baseline's near-zero R^2 , along with lower error metrics across the board.

```
--- Baseline Model Performance ---
MAE: 0.7077
MSE: 0.8573
RMSE: 0.9259
R2: -0.0001

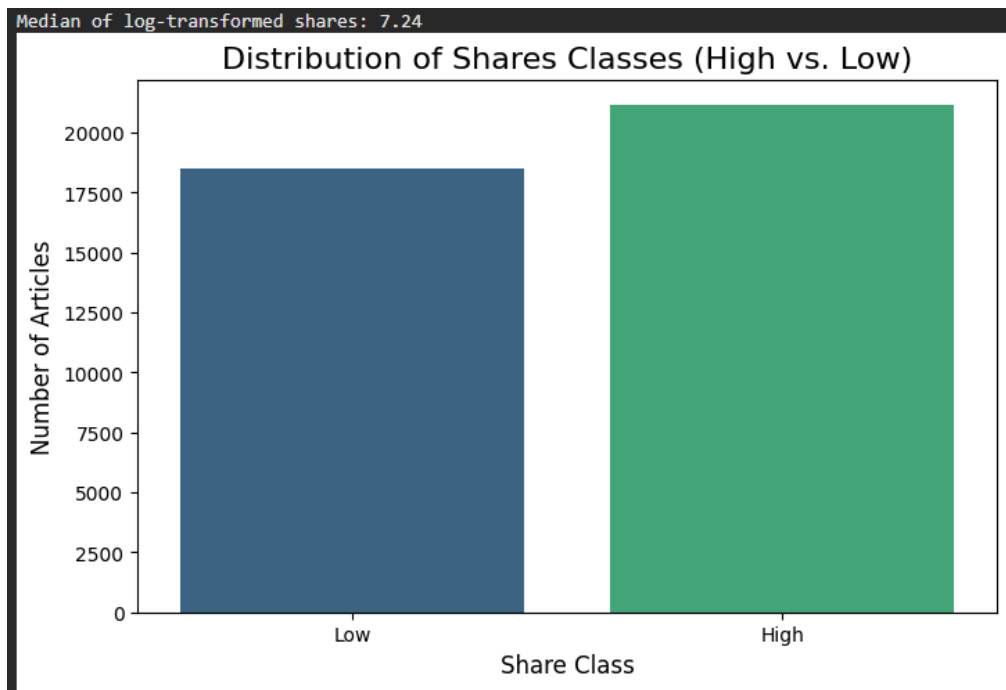
--- Linear Regression Model Performance ---
MAE: 0.6462
MSE: 0.7510
RMSE: 0.8666
R2: 0.1240

--- Comparison ---
Linear Regression (R2=0.1240) outperforms baseline (R2=-0.0001).
Improvement in R-squared: 0.1241
```

Classification Model

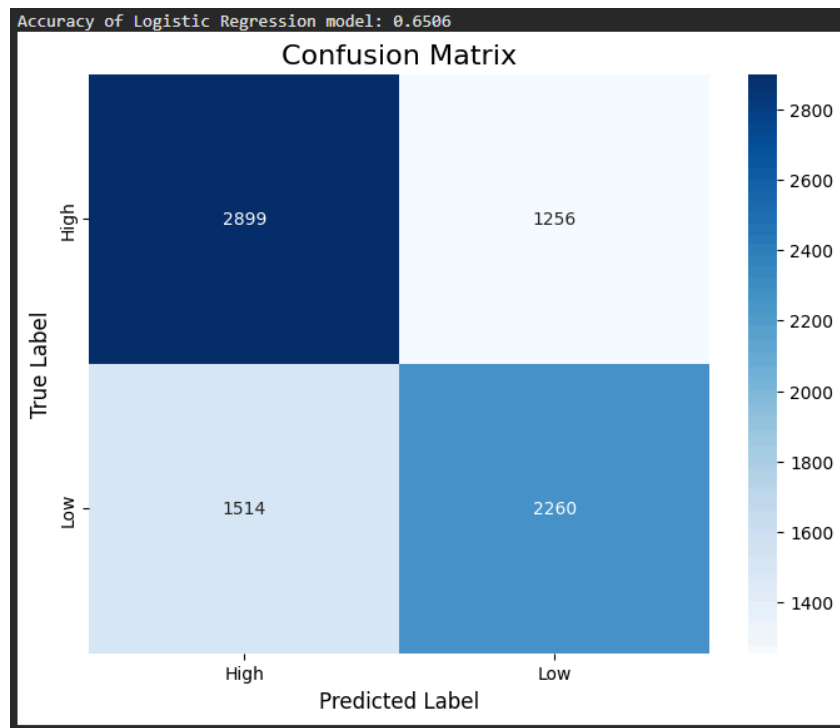
Classifying Articles as 'High' or 'Low' Popularity

For the classification task, `shares_log` was split into two categories based on its median value. Articles at or above the median were labeled "High", and those below were labeled "Low".



Model Training and Evaluation

A Logistic Regression model was trained to distinguish between the two classes. The same feature selection and scaling steps from the regression task were applied. Model performance was assessed using accuracy and a confusion matrix.



Interpreting the Coefficients

We analyzed the logistic regression coefficients to identify which features most strongly influenced the classification outcome. Features such as `kw_avg_avg`, `kw_max_avg`, and `data_channel_is_tech` stood out as particularly impactful.

Top 10 Most Influential Features (Logistic Regression):			
	Feature	Coefficient	Abs_Coefficient
23	kw_avg_avg	-0.962927	0.962927
22	kw_max_avg	0.567578	0.567578
13	data_channel_is_tech	-0.345646	0.345646
12	data_channel_is_socmed	-0.297649	0.297649
30	weekday_is_saturday	-0.249893	0.249893
31	weekday_is_sunday	-0.191146	0.191146
11	data_channel_is_bus	-0.187722	0.187722
24	self_reference_min_shares	-0.155777	0.155777
15	kw_min_min	-0.134441	0.134441
17	kw_avg_min	0.122435	0.122435

Hypothesis Testing on Selected Features

We ran hypothesis tests on five features to statistically assess their relationship with `shares_log`. For each feature, we defined null and alternative hypotheses and selected an appropriate test ; an independent samples t-test for binary features and Pearson Correlation for continuous ones.

data_channel_is_tech (t-test): Rejected H_0 . Tech articles showed a significantly different `shares_log` compared to non-tech articles.

```

H0: No significant difference in mean shares_log between tech and non-tech articles.
H1: There is a significant difference.

T-statistic: 12.0199
P-value:      4.2670e-33

Conclusion: Reject H0. Significant difference exists.

Mean shares_log -> Tech articles:      7.5820
Mean shares_log -> Non-tech articles: 7.4515

```

data_channel_is_socmed (t-test): Rejected H_0 . Social media articles differed significantly from non-social media articles.

```
H0: No significant difference in mean shares_log between socmed and non-socmed articles.  
H1: There is a significant difference.  
  
T-statistic: 17.7703  
P-value: 6.8817e-67  
  
Conclusion: Reject H0. Significant difference exists.  
  
Mean shares_log -> Socmed articles: 7.7781  
Mean shares_log -> Non-socmed articles: 7.4569
```

n_tokens_content (Pearson Correlation): Rejected H₀. A significant linear correlation with shares_log was found.

```
H0: No linear correlation between n_tokens_content and shares_log.  
H1: There is a linear correlation.  
  
Pearson Correlation Coefficient: 0.0248  
P-value: 7.5079e-07  
  
Conclusion: Reject H0. Significant linear correlation exists.
```

global_sentiment_polarity (Pearson Correlation): Rejected H₀. Sentiment polarity showed a significant relationship with shares.

```
H0: No linear correlation between global_sentiment_polarity and shares_log.  
H1: There is a linear correlation.  
  
Pearson Correlation Coefficient: 0.0489  
P-value: 1.8961e-22  
  
Conclusion: Reject H0. Significant linear correlation exists.
```

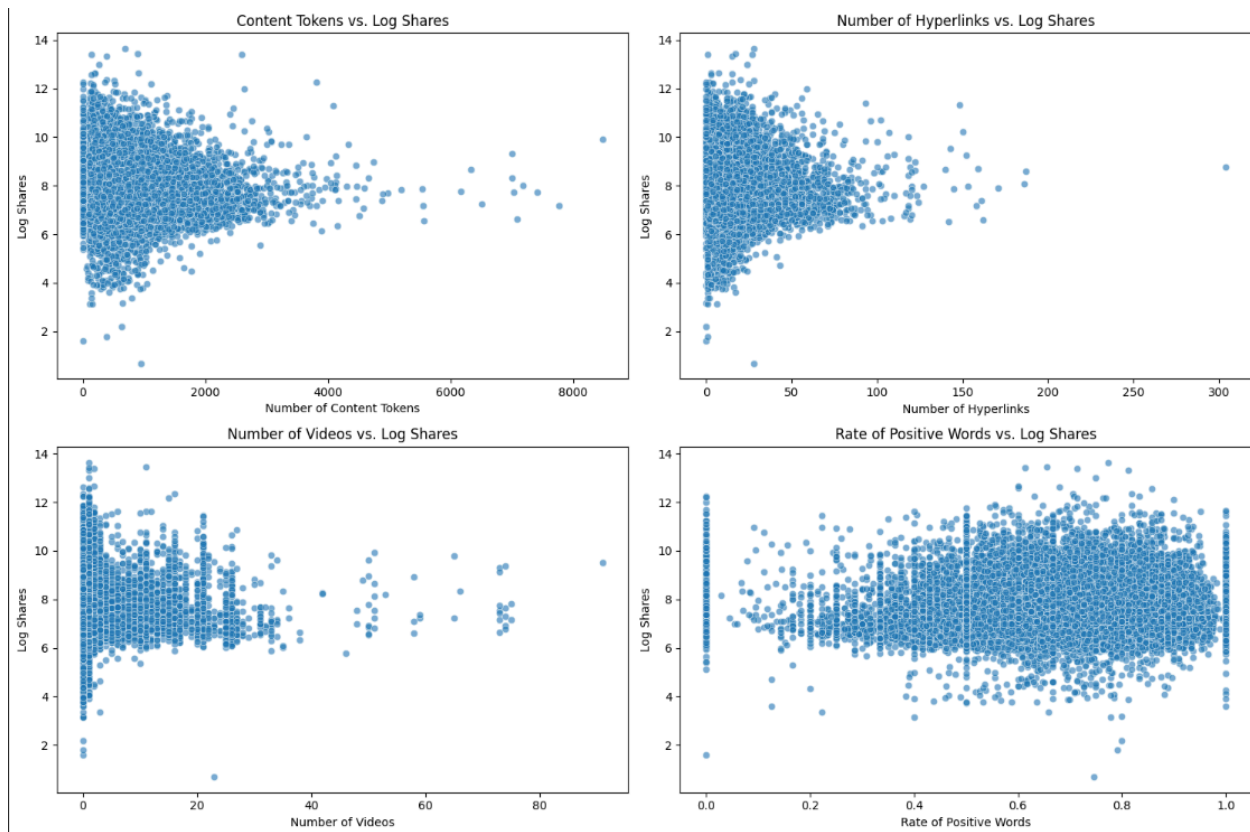
kw_avg_avg (Pearson Correlation): Rejected H₀. Average keyword performance was significantly correlated with shares.

```
H0: No linear correlation between kw_avg_avg and shares_log.  
H1: There is a linear correlation.  
  
Pearson Correlation Coefficient: 0.2218  
P-value: 0.0000e+00  
  
Conclusion: Reject H0. Significant linear correlation exists.
```

All five features demonstrated statistically significant relationships with `shares_log`, confirming their relevance to the prediction task.

Visualizing Relationships Between Key Features and the Target Variable

Scatter plots were created to visualize how selected features; including `n_tokens_content`, `num_hrefs`, `num_videos`, and `rate_positive_words`; relate to `shares_log`. These plots provided an intuitive, visual sense of each feature's individual influence.



Conclusion

The most significant predictors of article popularity are keyword-related features (especially `kw_avg_avg`), content channel (social media and tech articles outperform world news), and weekend publication timing. Sentiment polarity and content length, while statistically significant, have very weak practical effects. The fitted linear regression model meaningfully outperforms the mean baseline despite a modest R^2 , which is consistent with the inherently unpredictable nature of social media virality. Future work could explore ensemble models (Random Forest, Gradient Boosting) to capture non-linear relationships and improve predictive power.

FINAL SUMMARY: MOST SIGNIFICANT PREDICTORS OF ARTICLE POPULARITY

[Linear Regression: Top 5 by Absolute Coefficient]

	Coefficient
kw_avg_avg	0.444476
kw_max_avg	-0.262132
kw_avg_min	-0.083744
data_channel_is_tech	0.079018
average_token_length	-0.074168

[Logistic Regression: Top 5 by Absolute Coefficient]

Feature	Coefficient
kw_avg_avg	-0.962927
kw_max_avg	0.567578
data_channel_is_tech	-0.345646
data_channel_is_socmed	-0.297649
weekday_is_saturday	-0.249893

[Hypothesis Tests: All 5 Rejected H0 at alpha = 0.05]

Feature	Test	Result
data_channel_is_tech	t-test	Significant (p=4.27e-33)
data_channel_is_socmed	t-test	Significant (p=6.88e-67)
n_tokens_content	Pearson	Significant (p=7.51e-07)
global_sentiment_polarity	Pearson	Significant (p=1.90e-22)
kw_avg_avg	Pearson	Significant (p≈0.000)

KEY CONCLUSION:

kw_avg_avg is the single strongest predictor across both models.
Content channel (socmed, tech) and weekend timing also play
significant roles in determining article shareability.